Eftekhar Hossain

PhD Student, University of Central Florida

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Research Interest

My broad research interests are *Natural Language Processing* and *Multimodal Learning*. My work focuses on leveraging *Large Language Models* to understand how AI interprets complex information from multi-perspective narratives, aiming to enhance AI-driven comprehension across diverse viewpoints and modalities to support better decision-making in various applications.

Education

2024– **PhD., Computer Science**, *University of Central Florida*, USA.

2021–2023 **MSc., Electronics and Telecommunication Engineering**.
Chittagong University of Engineering and Technology, Bangladesh

Thesis : Investigating Multimodal Features to Identify Multilingual Offense from Social Media Memes
Thesis | *Slide*

2014–2018 **B.Sc., Electronics and Telecommunication Engineering**.
Chittagong University of Engineering and Technology, Bangladesh

CGPA : 3.88/4.00

Publications

Conference Proceedings

C11 Eftekhar Hossain, John Salvador, Santu Karmaker. ***"Evaluating Whether LLMs Can Reliably Connect the DOTs?"***. NeurIPS (Evaluations Track). 2026

C10 Omar Sharif, Eftekhar Hossain, Nikhil Singh, Patrick Ng. ***"Disentangling Perception and Reasoning in Multimodal LLMs via Reward Design"***. Findings of AACL. 2026

C9 Kawsar Ahmed, Md Osama, Omar Sharif, Eftekhar Hossain, Mohammed Moshiul Hoque. ***"Ben-NumEval: A Benchmark to Assess LLM's Numerical Reasoning Capabilities in Bengali"***. Findings of ACL. 2025

C8 Bowen Yan, Zhengsong Zhang, Liqiang Jing, Eftekhar Hossain, Xinya Du. ***"FIHA: Autonomous Hallucination Evaluation in Vision-Language Models with Davidson Scene Graphs"***. Findings of ACL. 2025

C7 Eftekhar Hossain, Sanjeev Kumar Sinha, Santu Karmaker *et al.* ***"LLMs as Meta-Reviewers' Assistants: A Case Study"***. NAACL. 2025

C6 Eftekhar Hossain, Omar Sharif, Mohammed Moshiul Hoque, and Sarah M Preum. ***"Deciphering Hate: Identifying Hateful Memes and Their Targets"***. ACL. 2024

C5 Shawly Ahsan*, Eftekhar Hossain*, Omar Sharif, Mohammed Moshiul Hoque *et al.* ***"A Multimodal Framework to Detect Target Aware Aggression in Memes"***. EACL. 2024

C4 Eftekhar Hossain*, Omar Sharif*, Mohammed Moshiul Hoque, and Sarah M Preum. ***"Align before Attend: Aligning Visual and Textual Features for Multimodal Hateful Content Detection"***. EACL-SRW. 2024

C3 Eftekhar Hossain, Omar Sharif, Mohammed Moshiul Hoque. ***"MUTE: A Multimodal Dataset for Detecting Hateful Memes"***. AACL-SRW. 2022

C2 Eftekhar Hossain, Omar Sharif, Mohammed Moshiul Hoque. ***"MemoSen: A Multimodal Dataset for Sentiment Analysis of Memes"***. Language Resources and Evaluation Conference (LREC). 2022

C1 Omar Sharif, Eftekhar Hossain, Mohammed Moshiul Hoque. ***"M-BAD: A Multilabel Dataset for Detecting Aggressive Texts and Their Targets"***. CONSTRAINT @ACL. 2022

Journal Articles

J4 Md Shalha Mucha Bhuyan, Eftekhar Hossain, Khaleda Akhter Sathi, Md Azad Hossain, M Ali Akber Dewan. ***"BVQA: Connecting Language and Vision Through Multimodal Attention for Open-Ended Question Answering"***. IEEE Access. 2025. **(Impact Factor: 3.367)**

J3 Ahatesham Bhuiyan, Eftekhar Hossain, Mohammed Moshiul Hoque, M Ali Akber Dewan. ***"Enhancing Image Caption Generation Through Context-Aware Attention Mechanism"***. Journal of Heliyon. Elsevier, 2024. **(Impact Factor: 3.4)**

J2 Eftekhar Hossain, Omar Sharif, Mohammed Moshiul Hoque, Md Azad Hossain *et al.*. ***"Identification of Multilingual Offense and Troll from Social Media Memes Using Weighted Ensemble of Multimodal Features"***. Journal of King Saud University - Computer and Information Sciences. Elsevier, 2022

J1 Eftekhar Hossain, Mohammed Moshiul Hoque, Enamul Hoque, Md Saiful Islam. ***"A Deep Attentive Multimodal Learning Approach for Disaster Identification From Social Media Posts"***. IEEE Access. 2022. **(Impact Factor: 3.367)**

Research Experience

Research Assistant, University of Central Florida

Aug 2024 – present ***Multi-Perspective Narrative Understanding using LLMs***.
(1) Developing a method for retrieving narratives that are similar with respect to a specific criterion. (2) Developing a framework to identify the overlap, unique, and conflicting information from pairs of alternative narratives.

Advisor : **Prof. Dr. Santu Karmaker**, *Bridge-AI Lab*, University of Central Florida (*Personal Web-page*)

Visiting Student (Remote), University of Texas at Dallas

Jan'24 – May'24 ***Evaluating Fine-grained Hallucination in Large Vision Language Models***.
Proposed an LLM- and annotation-free fine-grained evaluation framework for assessing object, attribute, and relationship hallucinations in LVLMs. *ACL*

Advisor : **Prof. Dr. Xinya Du**, *DuLab*, University of Texas at Dallas (*Personal Web-page*)

Visiting Student (Remote), Dartmouth College

May'23 – Dec'23 ***Multimodal Hateful Content Detection in Social Media***.
Proposed a context-aware and dual co-attention framework to meaningfully align the visual and textual features for multimodal hateful content detection. *EACL* | *ACL*

Advisor : **Prof. Dr. Sarah M. Preum**, *Persist Lab*, Dartmouth College (*Personal Web-page*)

Researcher, CUET NLP Lab, Bangladesh

July'19 – July'24 ***Multimodal Meme Understanding***.
Developed several benchmarks including MemoSen, MUTE, MIMOSA for Multimodal research in Bangla Language. *LREC* | *AACL* | *EACL*

Advisor : **Prof. Dr. Mohammed Moshiul Hoque**, , Chittagong University of Engineering and Technology, Bangladesh (*Personal Web-page*)

Professional Experience

Dec'22–July'24 **Assistant Professor, Dept. of Electronics and Telecommunication Engineering**.

Chittagong University of Engineering and Technology, Bangladesh

July'19–Dec'22 **Lecturer, Dept. of Electronics and Telecommunication Engineering**.
Chittagong University of Engineering and Technology, Bangladesh

Oct'18–June'19 **Lecturer, Dept. of Electrical and Electronic Engineering**.
Premier University, Chittagong, Bangladesh

Technical skills

Languages Python, C, C++, Shell

ML/DL PyTorch, TensorFlow, Keras, Scikit-Learn, Numpy, Pandas

LLM Libraries Huggingface, LangChain, LangGraph, vLLM, Ollama

Web Dev. Flask, Bootstrap, PHP, MySQL

IDE & Tools Jupyter, Spyder, VS Code, Git, LaTeX

Academic Achievements & Awards

2023 ***Best Research Award*** among Faculty of ECE at Chittagong University of Engineering and Technology (CUET), Bangladesh

2021 ***Winner*** in ***Hope Speech Detection Shared Task*** at EACL conference.

2nd ***Runner-up*** in ***Multimodal Troll Meme Detection and Offense classification Shared Task*** at EACL conference.

2017 ***1st Prize*** at the ***Project Show competition*** at CUET.

1st Prize at the ***Web Development Project Design Course*** at CUET.

2016 ***Merit Award*** for ***Obtaining Highest CGPA in an Undergraduate Academic Year*** at CUET.

Invited Training *Github Repo*

Feb 2024 ***Trainer*** of the short training on **"Machine Learning with Python"** Organized by *IT Bussiness Incubator at CUET.*

Feb 2023 ***Instructor*** of the course **"Hands on Training on Bangla Language Processing"** Organized under 2nd *International Workshop on Computational Linguistics and Bangla Language Processing (CLBLP)*, Bangladesh.

Oct 2022 ***Trainer*** of the short training on **"Machine Learning with Python"** Organized by *IT Bussiness Incubator at CUET.*

June 2022 ***Instructor*** of the one week long training on **"AI and Machine Learning with Python"** Organized by *IT Bussiness Incubator at CUET.*

Undergraduate Teaching

2023 **CSE 281: Data Structure and Algorithms**, CUET.
CSE 284: Object Oriented Programming, CUET.

2022 **CSE 181: Fundamentals of Computer Programming**, CUET.
CSE 284: Object Oriented Programming, CUET.

2021 **ETE 455: Neural and Fuzzy Systems**, CUET.
ETE 407: Multimedia Communication, CUET.

2020 **CSE 281: Data Structure and Algorithms**, CUET.
ETE 311: Information Theory and Coding, CUET.

ETE 317: Antenna and Wave Propagation, CUET.

2019 **EEE 201: Signal and System**, Premier University..

EEE 101: Electrical Circuits-I, Premier University.

2018 **EEE 435: Cellular Mobile Communication**, Premier University.

Co-Curricular

Judge and Organizer

Nov 2023 **Machine Learning Competition**, *ETE Infixon 2023*.

Feb 2023 **Natural Language Processing Competition 1**, *CLBLP 2023*.

Natural Language Processing Competition 2, *CLBLP 2023*.

Oct 2022 **Machine Learning Competition**, *ETE DecaHertz 2022*.

November 1, 2026